Algorithms 2

Vertex-connectivity in graphs

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1. Connectivity

1.1 Graph disconnectors

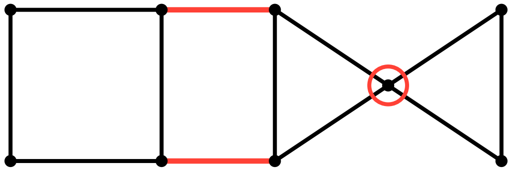
Definition 1.1.1

- For a graph G, write

$$C(G) := \text{number of connected components of } \; G$$

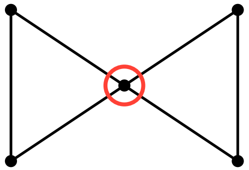
Definition 1.1.2

- Let $X \subseteq V(G) \cap E(G)$.
- If $C(G - X) > C(G)$, then X is called a *disconnector*



1.1 Graph disconnectors

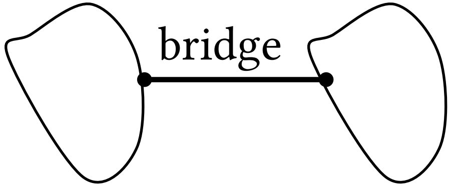
if $X \subseteq V(G)$, then X is a *vx-disconnector*.



- vx-disconnectors of size 1 are called *cut-vxs*.

- The ends of bridges are cut-vxs.
- The opposite is **not** true!

if $X \subseteq E(G)$, then X is a *edge-disconnector*.



- edge-disconnectors of size 1 are called *bridges*.

Observation 1.1.3 (H.W.)

$$e \in E(G) \text{ is a brige} \Leftrightarrow e \text{ does not lie on a cycle of } \; G$$

2. Whitney's Theorem

Theorem 2.1.1 (Whitney's Theorem)

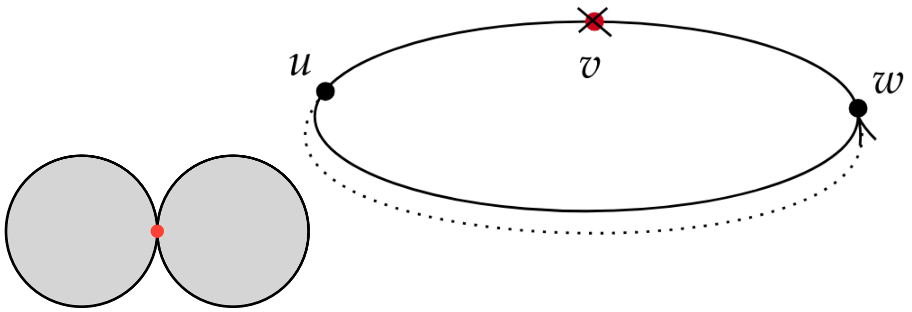
- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

proof $\Leftarrow$.

- Assume G has a cut vertex, denoted by v.
- v seperates the graph into 2 components C_1, C_2.
- Take $w \in C_1, u \in C_2$ both should hold true

1. A cycle $C := wx_1...x_\ell uw$ exists in G.
2. If we remove v then u and v are disconencted.



contridition: If $v \in C$ then removing v or any vertex in C wont seperate w from u, and if $v \notin C$ then the path $wx_1...x_\ell u$ still exists.

2.1 Whitney's Theorem

Theorem 2.1.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

$\Rightarrow$

- Suppose G has no cut-vxs.
- We prove by induction on

$$\delta_{G(u,v)} := \text{length of the shortest path u} \rightsquigarrow \text{v in } G$$

- Basis:
  - Let $u, v \in V(G)$ have $\delta_G(u, v) = 1$
  - If $uv \in E(G)$ is a brigde, the u, v are cut-vxs. **contridiction**.
  - From the observation uv lies on a cycle.

Observation 2.1.3 (H.W.)

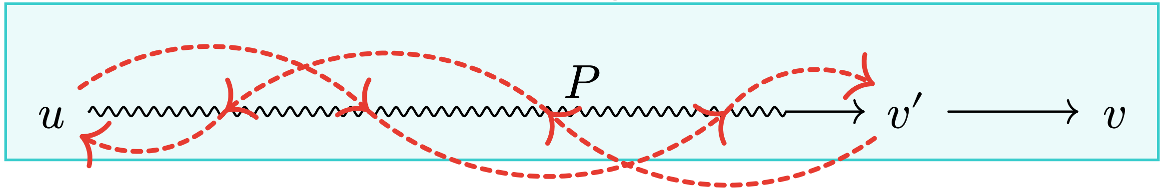
$e \in E(G)$ is a brige $\Leftrightarrow$ e does not lie on a cycle of G

Theorem 2.1.4 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

- Suppose G has no cut-vxs.
- Suppose that the claim hold true for all pairs of verecies u, v such that $\delta_{G(u,v)} < k$
- Consider a pair $u, v \in V(G)$ satisfying $\delta_G(u, v) = k$
- Let P be the shortest path $u \rightsquigarrow v$, and let v' denote the vertex before v on P.
- As $\delta_G(u, v') = k - 1$, there is a cycle C containting u, v'



- If $v \in V(C)$ we are done

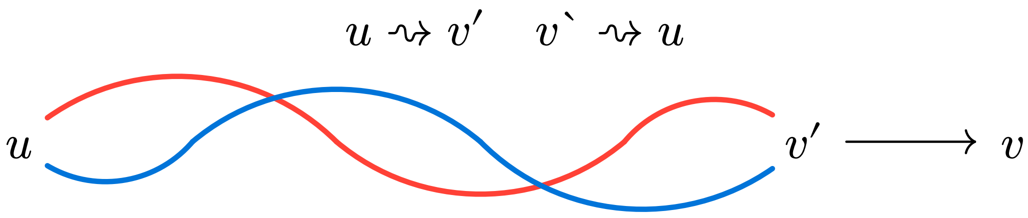
2.1 Whitney’s Theorem

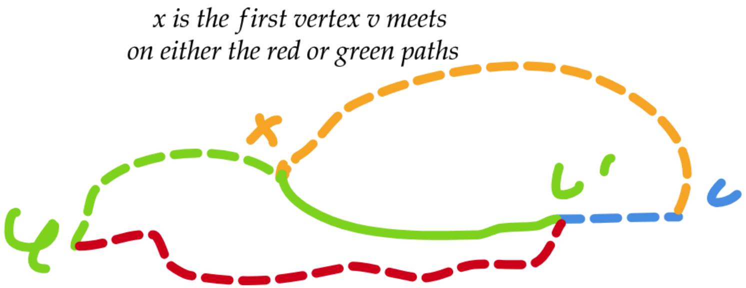
Theorem 2.1.5 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

- Otherwise, we can divide C into 2 arcs



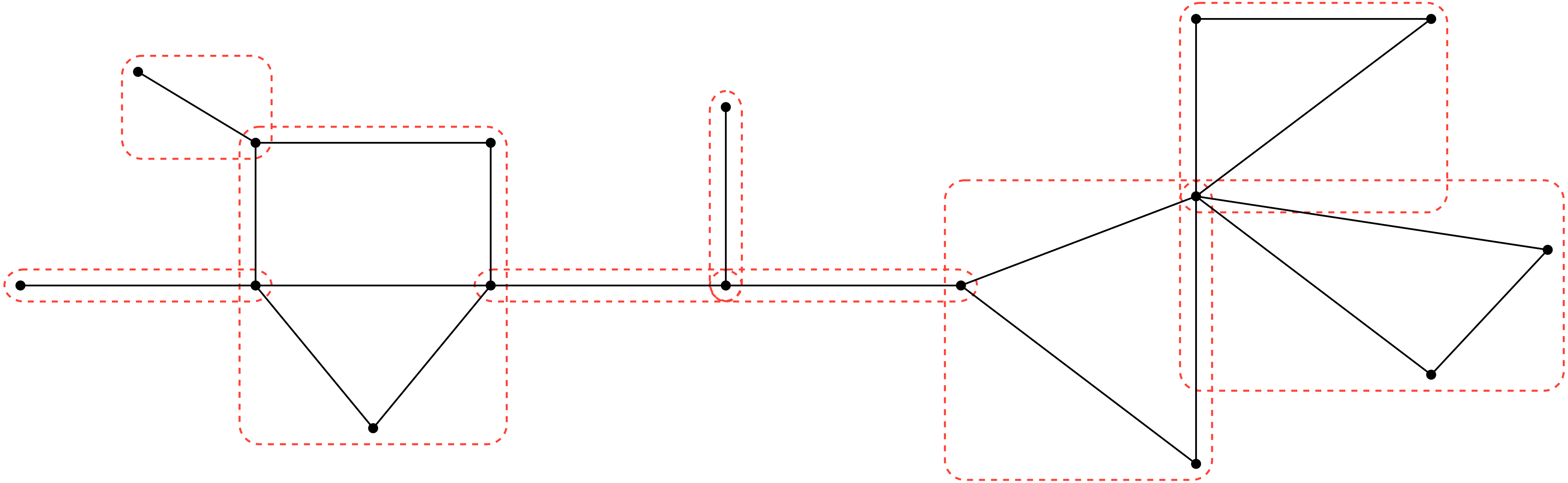


- As G has no cut-vxs, $G - v'$ is still connected,
- There is a path from v to a vertex x that lies on one of the arcs)
- W.L.O.G, let $x \in u \rightsquigarrow v'$ **take the shortest path from v to any of the arcs!**.
- Then the cycle $u \rightsquigarrow x \rightsquigarrow v \to v' \rightsquigarrow u$ is a cycle, and contains both u and v.

3. Consequences of Whitney's Theorem

3.1 Graph blocks

Definition 3.1.1 A maximal connected subgraph of G containing no cut-vxs is called a block of G



3.2 Consequences of Whitney's Theorem

Theorem 3.2.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } \ G \ \text{ lies on a common cycle.}$$

- Any two blocks of G meet in at most one vertex.
- If two block do meet, their intersection vertex is a cut-vertex.
- The blocks of G partition $E(G)$ (H.W.).
- Every cycle of G lies in precisly one block of G(H.W.).

3.3 Block tree

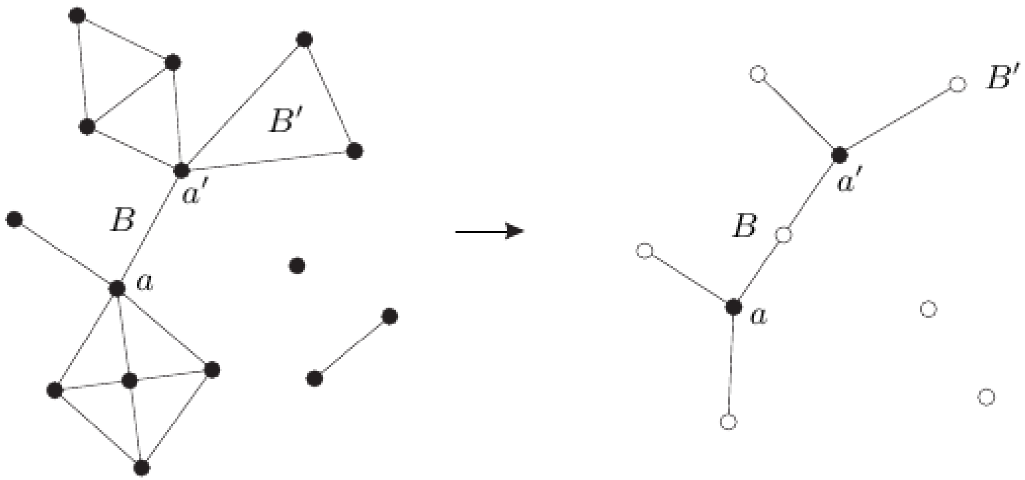
Given a graph G define the following auxiliary graph:

- $B(G) := \{B : B \subseteq G \text{ is a block of G}\}$
- $C(G) := \{v : v \in V(G) \text{ is a cut-vx of G}\}$

Let $\mathrm{BC}(G)$ denote the follwing graph

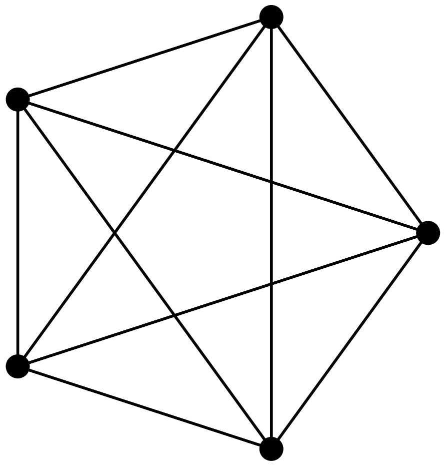
- The vertices of $\mathrm{BC}(G)$ are $B(G) \cap C(G)$
- The edges of $\mathrm{BC}(G)$ are
$$\{\{B, v\} : B \in B(G), v \in C(G) \text{ and } v \in B\}$$

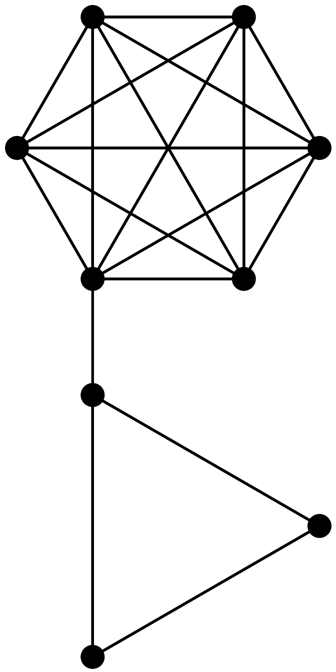
Theorem 3.3.1 if G is connected then $\mathrm{BC}(G)$ is a tree



4. Assesing connectivity

Question 4.1.1 Given these graphs, which one is more connected?

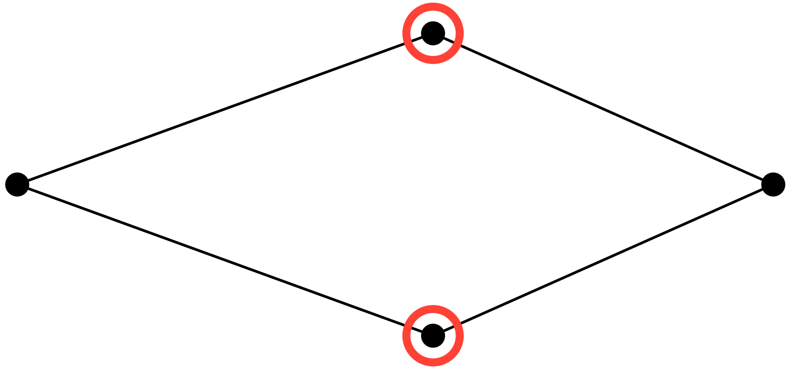




4.2 Definition I

Definition 4.2.1

- $0 \leq k \in \mathbb{N}$
- G - graph with $v(G) > k$
- If $G - X$ is conncted $\forall X \subseteq V(G), |X| \leq k - 1$ then G is called *k-connected*



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- If $G - X$ is conncted $\forall X \subseteq V(G), |X| \leq k - 1$ then G is called *k-connected*

- To be k-connected, $v(G) > k$ must hold
- What about K_1?
- We assume it is connected
- But our definition K_1 is not connected!
- We'll fix this later

4.3 Definition I

Definition 4.3.2 The largest $k \in \mathbb{Z}_{\geq 0}$ for which G is k-connected is denoted by $\kappa(G)$

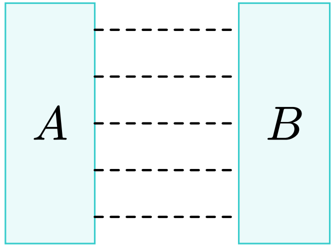
- All disconnected graphs have $\kappa(G) = 0$
- All connected graphs (but K_1) are **1-connected**
- $\kappa(G) \geq 1$
- $\kappa(K_1) = 0$
- Cycles with 3 or more vertecies are **2-connected**
- $\kappa(K_r) = r - 1$

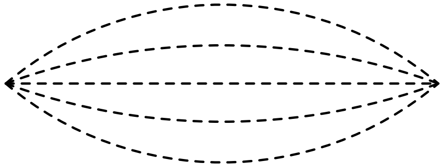
Again $\kappa(K_1) = 0$, is annoying, can we fix it?

4.4 Definition II: Path connectivity

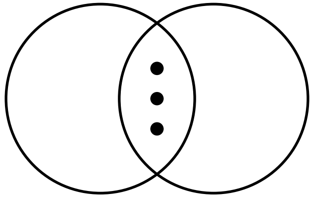
Definition 4.4.1

- G a graph
- $A, B \subseteq V(G)$
- let $\rho_G(A, B) :=$ the number of vx-disjoint (A, B)-paths





Here $|A| = |B| = 1$



$A \cap B \neq \emptyset$ also can happen

Remark

Vertices in A and B counts toward the vertices of the paths.
If $|A| = 1$ or $|B| = 1$ then we count the number of paths from the neibours of A or B respectively.

4.5 Definition II: Path connectivity

Definition 4.5.1 G is *k-connected* if $\rho_G(u,v) \geq k \quad \forall \{u,v\} \in \binom{V(G)}{2}$

So whats better: vertex connectivity or path connectivity ?

Menger's theorem tells us that they are the same!

4.6 Our definition

Definition 4.6.1 The minimum size of a vx set X such that $G - X$ is disconnected or has a single vx is called the *vx-connectivity* of G and is denoted by $\kappa(G)$

Definition 4.6.2 A graph G is called *k-connected* if $\kappa(G) \geq k$

Remark

Other than the complete graph:

$$\text{A graph is } k\text{-connected} \Leftrightarrow \text{all of its vx-cuts are of size} \geq k$$

The complete graph has no vx-cuts of any size, this is where the 2nd part of the definition comes in

5. Menger’s theorem

5.1 Menger's theorem

Definition 5.1.1

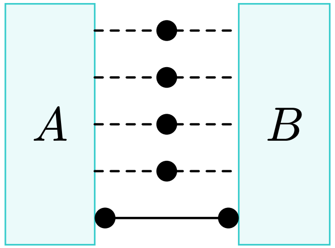
- $A, B \subseteq V(G)$
- Denote by $\kappa_G(A, B)$ the size of the **least** (A,B)-vx-disconnetor

Theorem 5.1.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

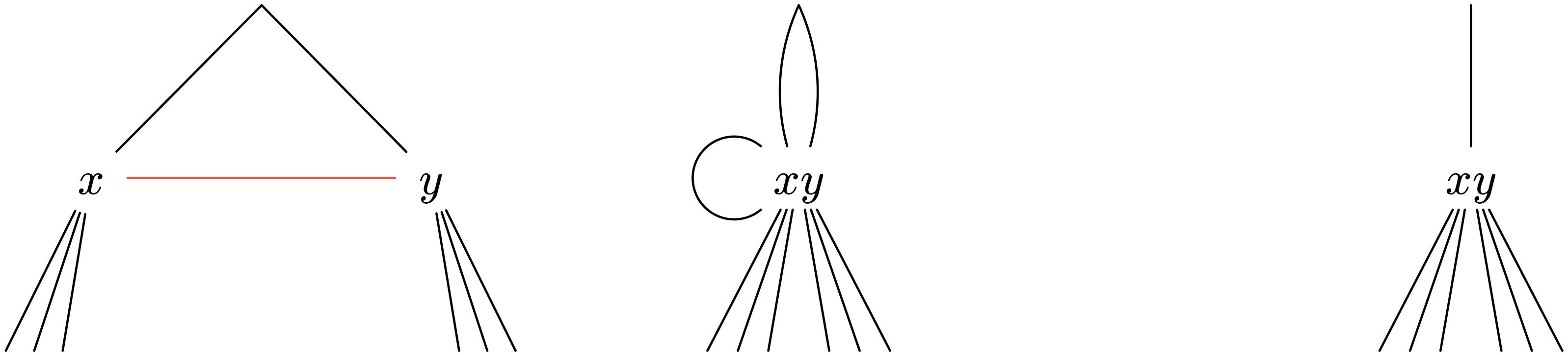
- You know this by the name "min-cut-max-flow"
- $\kappa_G(A, B) \geq \rho_G(A, B)$ is trivial
- We need to prove

$$\kappa_G(A, B) \leq \rho_G(A, B)$$



5.2 Edge contractions

- Consider the edge $e = xy$
- To constract e:
  - remove e
  - merge x and y into one vertex v_{xy}
  - connect v_{xy} to $N_G(x) \cap N_G(y)$, remove any dupplicant edges that may arise



- Denote by G/e the graph resualting from contracting e

5.3 Menger's theorem

Theorem 5.3.1 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

- Set $\kappa := \kappa_G(A, B)$ and $\rho := \rho_G(A, B)$
- Proof by induction on $e(G)$
- If $e(G) = 0$ the graph is disconnected
  - $\rho = |A \cap B| = \kappa$

5.3 Menger's theorem

Theorem 5.3.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

- assume $e(G) > 0$
- let $e = xy \in E(G)$ be an arbitrary edge
- In G/e, define $A_e \subseteq V(G/e)$ as follows
  - If $x, y \notin A$, then $A_e := A$
  - if $x \in A$,$y \notin A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
  - if $x \notin A$,$y \in A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
  - if $x, y \in A$, then $A_e := (A \setminus \{x, y\}) \cup \{V_e\}$
- Define B_e in the same manner

5.3 Menger's theorem

- Suppose that G/e has κ vx-disjoint (A_e, B_e)-paths
  - Any path not containing v_e exists in G
  - if none of the paths here contains v_e, then we are done
- suppose that there is a path in the linkage containing v_e.

  xy can be expanded in the path

  $$\begin{array}{ccccccccc} \text{---------} & \in N_G(x) & \text{———} & x & \text{———} & y & \text{———} & \in N_G(y) & \text{---------} \\ \text{---------} & \in N_G(y) & \text{———} & y & \text{———} & x & \text{———} & \in N_G(x) & \text{---------} \end{array}$$

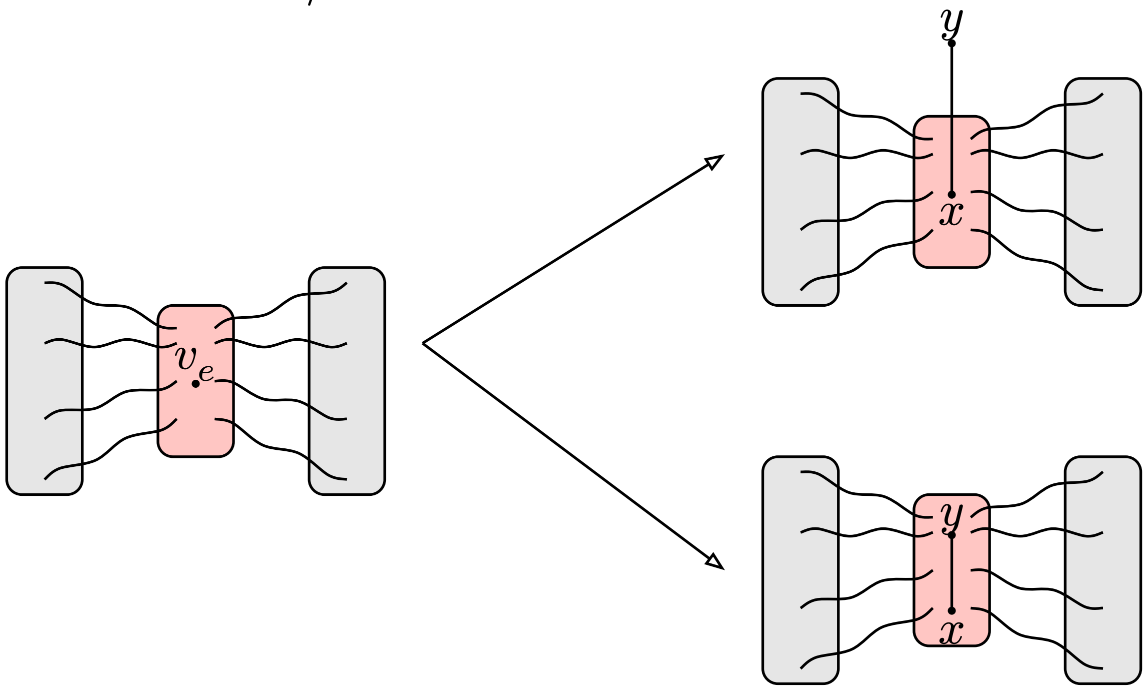
  xy cannot be expanded in the path

  $$\begin{array}{ccccccccc} \text{---------} & \in N_G(y) & \text{———} & x & \text{———} & y & \text{———} & \in N_G(y) & \text{---------} \\ \text{---------} & \in N_G(x) & \text{———} & x & \text{———} & y & \text{———} & \in N_G(x) & \text{---------} \end{array}$$

- In the second case, just take one of the verticies.

5.4 Menger's theorem

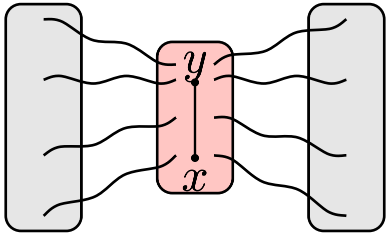
- Suppose that G/e don't have κ vx-disjoint (A_e, B_e)-paths
- $\kappa_{G_e}(A, B) < \kappa$
- Any (A_e, B_e)-vx-disconnector in G/e containe v_e
  - if not, G has an (A,B)-vx-disconnector of size $< \kappa$
- Any (A_e, B_e)-vx-disconnector in G/e is of size $\kappa - 1$



5.4 Menger's theorem

We are now in this position:

- $X :=$ (A, B)-vx-disconnector (Maybe $X = A$ or $X = B$)
- $|X| = \kappa$
- X spans e



- Note that by I.H.

$$\kappa_{G-e}(A, X) \geq \kappa \quad \text{and} \quad \kappa_{G-e}(X, B) \geq \kappa$$

5.5 Implication of Menger's theorem

Corollary 5.5.0.1 (H.W.)

- Let G be a graph. Then

$$\kappa(G) = \min\{\rho_G(u, v) : u, v \in V(G), u \neq v\}$$

Lemma 5.5.1 (The Extension Lemma H.W.)

- G - k-connected graph
- H obtained from G by:
  - Adding a new vertex
  - Connecting the new vertex to k arbitrary vertecies
- Then H is k-connected,

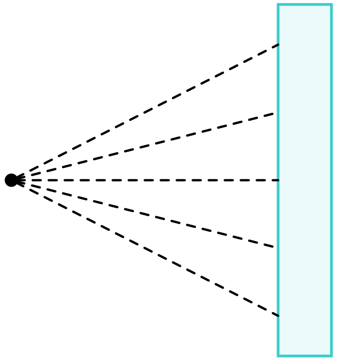
5.5 Implication of Menger's theorem

- Given $x \in V(G)$ and $B \subseteq V(G)$ s.t. $x \notin B$, we write *(x, B)-fan* to denote:

Lemma 5.5.2 (Fan Lemma H.W)

- G - k-connected graph
- $x \in V(G)$ and $B \subseteq V(G) \setminus \{x\}$

Then G has an (x, B)-fan if size $\min\{k, |B|\}$



5.6 Dirac's theorem

Theorem 5.6.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k-connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constains S

- assume $k \geq 3$
- let $x \in S$ be arbitrary and set $T := S \setminus \{x\}$
- $\kappa(G - x) \geq k - 1$ So there is a cycle C in $G - x$ containing T.
- let F be an (x, C)-fan in G of size $\min\{k, v(C)\}$
- Note the T partitions C into $k - 1$ arcs
- At least two of the paths in the fans land in the same arc on C
- We can extend C to conatin x